MIL-C-48845(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

CONNECTOR, RECEPTACLE, ELECTRICAL, for M5 BAROSWITCH

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for three types of non-moisture resistant miniature electrical connectors entitled Connector Receptacle Electrical which are used on the Nike Hercules warhead section.

1.2 Classification. This specification covers electrical connectors of the following types, sizes, and styles as specified.

1.2.1 Types. Square flange box mounting, solder cup type, contact and pin type for printed circuit use.

1.2.2 Sizes. The sizes shall be as specified on the detail drawings.

1.2.3 Arrangements. Arrangement of contacts shall be as shown on detail drawings.

1.2.4 Styles. Style S - Inserts containing socket contacts.
Style P - Inserts containing pin contacts.

1.2.5 Service rating. 7-1/2 amperes, 250 volts dc.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-267 - Aluminum Alloy, 24S, Bars, Rods, and Shapes Extruded.
QQ-A-268 - Aluminum 24S, Bars, Rods, and Wire Rolled or drawn.
QQ-A-325 - Aluminum Alloy 61S, Bars, Rods, Shapes, and Wire Rolled or Drawn.
QQ-A-351 - Aluminum 17S, Bars, Rods, and Wire Rolled or drawn.
QQ-A-416 - Plating, Cadmium (Electric deposited).
QQ-A-591 - Aluminum Alloy, Die Castings.
WW-T-785 - Tubing, Aluminum Alloy 24S, Round, Square, Rectangular, and other Shapes, Seamless, and Drawn.
WW-T-789 - Tubing, Aluminum Alloy, 61S, Round, Square, Rectangular, and other Shapes, Seamless, and Drawn.

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SPECIFICATIONS

MILITARY

MIL-M-14	- Molding Plastics and Molded Parts, Thermosetting.
MIL-G-2550	- General Specification for Ammunition, Except Small Arms Ammunition.
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-202	- Test Methods for Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8803876	- Connector, Receptacle, Electrical (Special 20 Pin).
8805851	- Specification for Connector, Receptacle, Electrical.
EL-8816849	- Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.1.1 Shells. Shells may be machined or die-cast. Die-cast parts shall conform to composition No. 1, 2, 5, 5A, 10, or 11 of QQ-A-591. Machined parts shall conform to composition 24S aluminum alloy per QQ-A-268, 61S aluminum alloy per QQ-A-325, 17S aluminum alloy per QQ-A-351, 24S aluminum alloy per WW-T-785, or 61S aluminum alloy per WW-T-789.

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3.1.2 Insert material. The insert material shall be glass fiber filled diallyl phthalate resin in accordance with MIL-M-14.

3.1.3 Contacts. Contacts shall be made from a suitable conductive material. All contacting surfaces shall be smooth and shall be silver plated 0.00015 inch minimum thickness followed by gold plate 0.00002 inch minimum thickness.

3.1.4 Finish. All metal parts, other than electrical contacts, shall be cadmium plated per QQ-P-416, Type II, Class C, except that a preliminary plating of other metal is permissable.

3.1.5 "O" rings. "O" rings shall be silicone rubber, 70 Shore A durometer hardness and be positioned in a groove in the receptacle. The "O" ring shall be retained in position so that it will not become dislodged during normal service operation.

3.2 Assembly. The connector shall comply with all requirements specified on the drawing (dwg) 8805851, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.2.1 Contact removal. Contacts need not be removable for soldering purposes. Pin contacts may be rigidly mounted in their inserts. All socket contacts shall be designed so that they can float easily.

3.2.2 Point of contact engagement. The distance from the front of the shell to a line representing the point which a square ended pin at mean diameter starts to engage the spring member of the socket shall be 0.260 inch maximum.

3.2.3 Insert construction. Inserts shall be constructed with proper sections and radii in order that they will not readily chip, crack, or break in assembly, or in normal service. Hollow type inserts shall not be used. Depressions on front and rear pin insert disks and rear socket insert disks, used to achieve creepage distance, shall not be located directly opposite on front and rear surfaces of individual disks when the resultant reduction in cross-sectional area would cause structural weakness. To prevent cross plugging of alternate positions, socket insert front face contact entrance holes and chambers, if used, shall be as small as practicable. Socket inserts shall provide suitable protection against a socket containing a pin before the plug has been polarized with its receptacle.

3.2.3.1 Removability of inserts. Inserts for plugs and receptacles need not be removable.

3.2.3.2 Insert positioning. Positive provision shall be made to position all inserts to prevent rotation with respect to shell. Over-all side play, or the difference between insert keyway and shell key dimensions, shall be 0.010 inch maximum.

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3.2.3.2.1 Alternate insert positioning. Insert positions shall be in accordance with dwg 8805851, and provided with alternate position slots. The alternate position slots shall be shorter than the insert so that the insert cannot be placed in an alternate position without first removing this obstruction. Unless otherwise specified by the applicable drawing or procuring agency, connectors shall be furnished with the insert in the normal (N) position. Connectors furnished in an alternate position may have a full length normal (N) positioning slot.

3.2.3.3 Contact arrangement. Contacts shall be arranged in accordance with the applicable detail drawings.

3.2.3.4 Contact alignment. To facilitate self-alignment of mating contacts, all sockets for socket contacts shall be so designed that individual contacts will have over-all side play of 0.005 inch minimum in both the front and rear.

3.2.4 Snap rings. Snap rings used to assemble the insert within the shell shall not reduce the voltage breakdown nor move out of position during service. Snap rings shall be square or rectangular in cross section.

3.2.5 Polarization. Polarization of the plug with its receptacle shall be accomplished by an integral key, and shall be of such design that it cannot possibly interfere with the functions of the coupling threads. The integral key shall be of sufficient size in order that in normal use (without using excessive force) the connectors can be mated in only one position. The polarization of the plug with its receptacle shall be accomplished before engaging the coupling ring.

3.2.6 Ease of mating. Plugs and receptacles shall be capable of being fully mated and unmated by hand without the use of tools.

3.2.7 Screw threads. All coupling screw threads shall be as specified on dwg 8805851. Screw threads shall be checked with ring gages only. Slight out of roundness is not objectionable if the threads can be checked without forcing of the thread gages. Screw threads may be relieved provided such relief does not interfere with proper performance of the screw threads.

3.2.8 Interchangeability. All connectors bearing the same identification shall be directly and completely interchangeable with each other with respect to installation and performance. Removable inserts shall be interchangeable between shells of the same type and size.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

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3.5 Indentification of product. Each connector shall be clearly marked by metal stamping, before plating, on the connector shell with the supplier's name or trademark and part number, insert position, and the procuring agency's number followed by the appropriate detail dwg number. If insufficient space exists on the connector for the supplier's name or trademark and part number, insert position, and procuring agency's number, the supplier's part number may be omitted.

3.5.1 Insert position. One of the following letters, N, V, W, X, Y, or Z, as applicable, shall be added to the supplier's part number after the style letter.

3.5.2 Marking of inserts. Inserts shall be marked as shown on the applicable detail drawings.

3.5.2.1 Contact designation. Contacts shall be designated as shown on the applicable detail drawings. Letters shall be clearly legible, either raised or depressed, on the front and rear surfaces of each pin and socket insert, wherever a 1/16 inch minimum unobstructed surface exists adjacent to the contact. Arrangements shall be such as to avoid confusion between contacts. Lettering of socket inserts shall correspond with that of the mating pin insert.

3.6 Performance.

3.6.1 High potential. The connectors shall show no evidence of breakdown when 1500 volts dc is applied for one minute between any pair of contacts and between any contact and shell when tested as specified in 4.4.1.

3.6.2 Resistance of contacts. The resistance of pin and socket contacts when assembled as in service shall be such that the potential drop at 7-1/2 amperes shall be 23 millivolts maximum when tested as specified in 4.4.2.

3.6.3 Insulation resistance. The insulation resistance shall be 5000 megohms minimum when measured separately between any pair of contacts and between any contact and shell when tested as specified in 4.4.3.

3.6.4 Contact engaging and separating forces. Socket contacts in connectors shall meet the minimum and maximum engaging and separating force requirement of 1/2 ounce minimum and 12 ounces maximum when tested as specified in 4.4.4.

3.7 Environmental testing. The connectors shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The connectors shall also comply with applicable paragraphs in 3 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

3.7.1 Vibration. Complete connector assemblies shall show no evidence of damage such as cracks, chips, or loosening of components when tested as specified in 4.4.5.1.

3.7.2 Durability. Connectors shall be capable of being mated and shall show no mechanical or electrical defects after subjection to 500 cycles of insertions and withdrawals at a rate of 600 cycles per hour minimum when tested as specified in 4.4.5.2.

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3.7.3 Impact strength. Connectors containing socket contacts shall withstand the specified physical impact without cracking or chipping when tested as specified in 4.4.5.3.

3.7.4 Temperature cycling. Unmated connectors shall withstand exposure to five cycles of minimum temperature changes as indicated in Table I when tested as specified in 4.4.5.4.

Table I

Sequence	°C	°F
Start (HIGH)	+85 ± 2.5	+185
Reduce (LOW)	-55 ± 2.5	-67
Return to	+85 ± 2.5	+185

3.7.5 High potential, 50,000 feet altitude. The connectors shall show no evidence of breakdown when tested as specified in 4.4.5.5.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 10 complete assemblies (see dwg 8805851) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests need not be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. High potential (see 3.6.1)		X
3. Resistance of contacts (see 3.6.2)		X
4. Insulation resistance (see 3.6.3)		X
5. Contact engaging and separing forces (see 3.6.4)	X	
6. Vibration (see 3.7.1)	X	
7. Durability (see 3.7.2)	X	
8. Impact strength strength (see 3.7.3)	X	
9. Temperature cycling (see 3.7.4)	X	
10. High potential-altitude (see 3.7.5)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

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Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	293
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Assembly (see dwgs 8803876 and 8805851).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Major diameter of external thread (max.)	Gage
102.	Location of mounting holes incorrect	Gage
*103.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
**104.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent pins)	Visual
105.	Burred, bent, or misaligned contact (use acceptable mating part for gaging)	Gage
106.	Improper plating on any contact causing exposure of base metal such as peeling, flaking, chipping, etc	Visual
107.	Insufficient tinning or solder on inside of any solder pot	Visual
108.	Crack or void in shell or insert	Visual
109.	Insert improperly located with reference to polarizing keyway	Visual

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110.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Incorrect length to any contact point (0.164 to 0.250)	Gage
202.	Insufficient or improper plating on shell exposing base metal	Visual
203.	Scratches or abrasions on shell causing exposure of base metal	Visual
204.	Plastic flash on any contact	Visual
205.	Foreign matter between contacts or any contact and shell such as dirt, bits of wire, or any other conductive type matter	Visual
206.	Evidence of poor workmanship (see 3.8)	Visual
207.	Serialization improper or missing (see 3.4)	Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4.1) shall be used to determine that variation from the normal height. Reject any pin that varies by more than 0.063 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent sockets, Gage (see 4.3.4.1) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector insert shall be utilized as the reference surface for all measurements of socket depths.

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4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing and packaging not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	High potential	Major
b.	Resistance of contacts	Major
c.	Insulation resistance	Major
d.	Contact engaging and separating force	Major
e.	Vibration	Major
f.	Durability	Major
g.	Impact strength	Major

Of these tests, (d) through (g) are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 High potential, resistance of contacts and insulation resistance. Each connector shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.6.1, 3.6.2 and 3.6.3, the item shall be rejected and removed from the lot.

4.3.3.2 Contact engaging and separating forces, vibration, durability and impact strength. A sample shall be selected at random in accordance with Table V. The sample shall be tested in accordance with 4.4.4, 4.4.5.1, 4.4.5.2 and 4.4.5.3. If any item fails to comply with 3.6.4, 3.7.1, 3.7.2 and 3.7.3, the lot shall be rejected.

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Table V

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
2-8	1
9-15	2
16-25	3
26-40	5
41-65	8
66-110	13
111-180	22
181-300	36
300-and over	MIL-STD-105 (AQL 0.25 percent)

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8816849). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 High potential. The connectors shall show no evidence of breakdown when 1500 volts dc is applied for one minute between any pair of contacts and between any contact and shell. The voltage shall be applied gradually at the rate of 500 volts each second until 1500 volts is reached.

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4.4.2 Resistance of contacts. The resistance of pin and socket contacts when mated as in service shall be such that the potential drop at 7-1/2 amperes shall be 23 millivolts maximum. The electrical resistance shall be determined by measuring the potential drop across the contacts of the assembled connectors when carrying the specified current. The potential drop shall be measured at the extreme end of the solder cups. Each reading in millivolts shall be observed and recorded.

4.4.3 Insulation resistance. The insulation resistance shall be 5,000 megohms minimum when measured separately between any pair of contacts, and between shell and any contact. All unenergized connector pins shall be grounded to the shell. Measurements shall be made using a potential of approximately 500 ± 50 volts DC. Electrification time shall be two minutes maximum. Each reading in megohms shall be observed and recorded.

4.4.4 Contact engaging and separating forces. Using a steel pin of minimum diameter with hemispherical tip, with a 12 ± 4 microinch finish, socket contacts in connectors under test shall meet the minimum and maximum engaging and separating force requirement of 1/2 ounce minimum and 12 ounces maximum. Readings shall be taken with gradually increasing loads until the pin either engages within or separates from the socket contact.

4.4.5 Environmental testing (see 3.7).

4.4.5.1 Vibration. Complete connector assemblies shall be vibrated in accordance with procedure I of MIL-E-5272. The cable shall be rigidly attached at a point one foot from the connector with a small amount of slack in the cable. After vibration the coupling ring or other parts shall not loosen. The connectors shall not be held together with lock wires for this test. See figure 1 for typical vibration test setup.

4.4.5.2 Durability. The assembly of mating connectors, less the coupling ring used for the resistance of contacts test, shall be subjected to 500 cycles of insertions and withdrawals at a rate of 600 cycles per hour minimum. The insertions and withdrawals shall be accomplished in a manner similar to that which the connector shall be subjected to in service. After 500 cycles, the plug and receptacle shall show no mechanical or electrical defects and shall be tested as specified in 4.4.2 except that the contact resistance shall be 24 milliohms and the test current shall be 3 amperes.

4.4.5.3 Impact strength. Connectors containing socket contacts shall withstand, without cracking or chipping, the physical impact of a 66.5 ± 2 gram, one-inch solid steel ball dropped upon the front face of the insert from a free height of 18 inches minimum.

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4.4.5.4 Temperature cycling. Unmated connectors shall be exposed to at least five cycles of temperature changes as indicated in Table I, paragraph 3.7.4. The connectors shall be held at each specified temperature, for not less than 30 minutes. The connectors shall be transferred from one chamber to another for these changes and shall not be kept at room temperature for more than two minutes between changes. At the end of this test, the insert shall not crack or rotate in its shells. Within 45 minutes after this test, the connectors shall meet the high potential test of 4.4.1.

4.4.5.5 High potential at 50,000 feet altitude. The connector shall show no evidence of breakdown at 50,000 feet of altitude minimum when 450 volts dc minimum is applied as in 4.4.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The connectors shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The connectors shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The connectors shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

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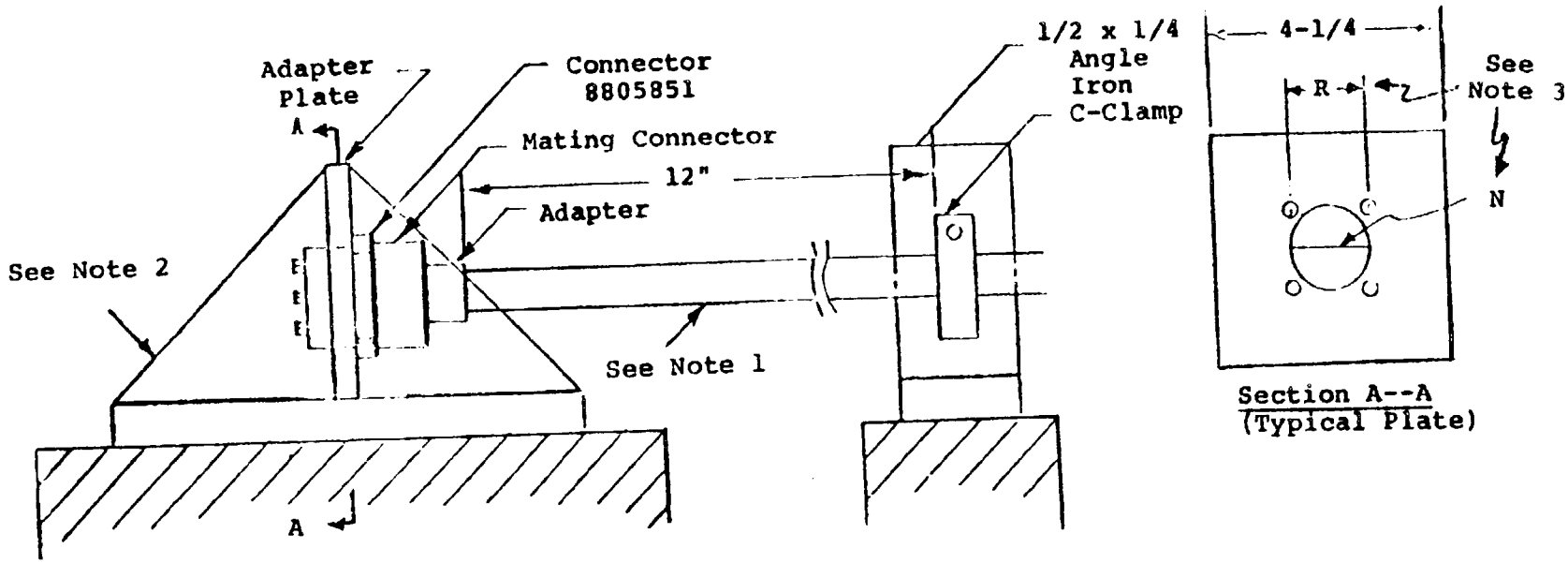
6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1163 (Rev. 1) dated 28 September 1962.

Custodian:
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Preparing activity:
Army - PA

Project No. NUOR-A041

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NOTES:

1. AN wire size corresponding to contact size and bundled.
2. Mounting bracket 1/4" welded steel construction.
3. R and N per applicable detail drawing.

Figure 1. Typical Vibration Test Setup

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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